

Shuaicheng Zhang

PHD CANDIDATE · COMPUTER SCIENCE DEPARTMENT OF VIRGINIA TECH

✉ zshuai8@vt.edu | 🏠 <https://zshuai8.github.io> | 🌐 <https://github.com/zshuai8>

Research Interest

My research focuses on **open world graph learning** which aims to develop graph machine learning models that operate effectively on real world evolving, heterogeneous graph data through unified theoretical and algorithmic innovations, enabling capability of reasoning over complex relational data at scale and significantly enhance decision-making in critical applications. My recent research interests focus on advancing graph learning methods that address temporal dynamics, structural heterogeneity, and heterophilic relationships, with applications spanning financial systems, industrial predictive maintenance, federated learning, temporal reasoning, and knowledge representation.

Education

Virginia Tech

PHD CANDIDATE COMPUTER SCIENCE

- Advisor: Dr. Dawei Zhou

Blacksburg, VA

August 2020 - present

MIT CSAIL

VISITING PHD STUDENT

- Advisor: Dr. Julian Shun

Cambridge, MA

May 2023 - Aug 2023

University of Maryland, College Park

MS TELECOMMUNICATIONS

- Advisor: Dr. Michael Dellomo

College Park, MD

August 2018 - May 2020

Virginia Tech

BS COMPUTER SCIENCE

- Minors in Mathematics

Blacksburg, VA

August 2013 - Dec 2017

Industrial Experience

2025 **Research Intern, Microsoft Research**, Redmond, WA, USA

2023 **Research Intern, MIT-IBM lab**, Cambridge, MA, USA

2021 **NLP Research Intern, Nature Language Processing Team**, Deloitte, NY, USA

2017 **Software Development Engineer Intern, Trust Team**, Hundsun Technologies Inc., Hangzhou, China

Preprints & Publications

Scientific Hypothesis Generation and Validation: Methods, Datasets, and Future Directions —Adithya Kulkarni, Fatimah Alotaibi, Xinyue Zeng, Longfeng Wu, Tong Zeng, Barry Menglong Yao, Minqian Liu, **Shuaicheng Zhang**, Lifu Huang, Dawei Zhou. In Arxiv, Under review

CAPTAIN: Conformal-Prediction-Based Multi-Source Time-Series Forecasting —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. Under review of **TMLR**'2025.

PDMBench: A Standardized and Multi-Perspective Benchmark for Predictive Maintenance on Multi-Sensor Industrial Time-Series Data —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. Under review of **ICLR**'2025.

HeroFilter: Adaptive Spectral Graph Filter for Varying Heterophilic Relations —**Shuaicheng Zhang**, Haohui Wang, Junhong Ling, Xiaojie Guo, Yada Zhu, Si Zhang, Dongqi Fu, Dawei Zhou. (*Equal Contribution) In proceedings of the (NeurIPS)'2025.

When Heterophily Meets Heterogeneity: New Graph Benchmarks and Effective Methods —Junhong Lin, Xiaojie Guo, **Shuaicheng Zhang**, Yada Zhu, Julian Shun. In Preceedings of the ACM KDD'2025. **Best Paper Award**

MentorPDM: Learning Data-Driven Curriculum for Multi-Modal Predictive Maintenance —**Shuaicheng Zhang**, Tuo Wang, Stephen Adams, Sanmitra Bhattacharya, Sunil Reddy Tiyyagura, Edward Bowen, Balaji Veeramani, Dawei Zhou. In Preceedings of the ACM KDD'2025.

UnifiedGT: Towards a Universal Framework of Transformers in Large-Scale Graph Learning —Junhong Lin, Xiaojie Guo, **Shuaicheng Zhang**, Yada Zhu, Dawei Zhou, Julian Shun. In Preceedings of the IEEE BigData 2025. **Patent Pending**.

TGEditor: Task-Guided Graph Editing for Augmenting Temporal Financial Transaction Networks —**Shuaicheng Zhang**, Yada Zhu, Dawei Zhou. In Proceedings of the ACM International Conference on AI in Finance(ICAIF)'2023(oral). **Patent Pending, Deployed at Wellsfargo**.

Personalized Federated Learning under Mixture of Distributions —**Shuaicheng Zhang***, Yue Wu*, Yanchi Liu, Quanquan Gu, Dawei Zhou, Haifeng Chen, Wei Cheng. (*Equal Contribution) In Proceedings of International Conference on Machine Learning(ICML)'2023.

Extracting Temporal Event Relation with Syntax-guided Graph Transformer —**Shuaicheng Zhang**, Ning Qiang, Lifu Huang. In Findings of North American Chapter of the Association for Computational Linguistics(NAAACL)'2022.

Awards & Honors

- 2025 **KDD Best Paper Award in Benchmark and Dataset Track**, KDD 2025
- 2025 **KDD Outstanding Reviewer (Top 10%)**, KDD 2025
- 2024 **CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
- 2023 **CCI SWVA Cyber Innovation Scholars**, Commonwealth Cyber Initiative (CCI) Southwest Virginia
- 2022 **CIKM Travel Award**, CIKM 2022
- 2021 **Virginia Tech Pratt Fellowship**, Virginia Tech Department of Computer Science

Academic Service

CONFERENCE REVIEWER

- 2026 **AAAI 2026, ICLR2026**, Reviewer, Program Committee
- 2025 **KDD 2025, ICML 2025, ICLR 2025, Neurips 2025**, Reviewer, Program Committee
- 2024 **KDD 2024, ICML 2024**, Reviewer, Program Committee
- 2023 **KDD 2023, ICDM 2023, WWW 2023**, Reviewer
- 2022 **SIGIR 2022, KDD 2022, NLPCC 2022, WWW 2023**, Reviewer
- 2021 **SDM 2022, WWW 2022, CIKM 2022**, Subreviewer
- 2021 **NLPCC 2021**, Reviewer

ORGANIZING COMMITTEE

- 2022 **The 1st Workshop on Trustworthy Learning on Graphs (TrustLOG) at The Conference on Information and Knowledge Management (CIKM)'2022**, Publicity Chair Atlanta

Skills

STATISTICS Advanced statistical analysis, inference, factor, regression analyses, hypothesis testing, hierarchical modeling

MACHINE LEARNING AND DEEP LEARNING TensorFlow, Keras, Pytorch, Scikit-Learn, XGBoost, Pandas, Numpy, Scipy, NLTK, NetworkX

PROGRAMMING Python, Java, C, Matlab, Shell, MySQL, HTML

BASICS Linux, Git, LaTeX, Docker

Teaching Experience

- Spring 22 **CS 5984 Computational Social Science**, Teaching Assistant
- Fall 21 **CS 5984 Natural Language Processing**, Teaching Assistant
- Spring 21 **CS 2505 Computer Architecture I**, Teaching Assistant
- Fall 20 **CS 3214 Operating System**, Teaching Assistant